

poisson_binomial

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Poisson-binomial using daily historical (Strike in Percent)

1 Import Packages and Set File Paths

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

FISHING_PATH = "../data/stx_com_csl_daily_8623.csv"
WIND_PATH = "../data/st_croix_wind_data.csv"
WAVE_PATH = "../data/st_croix_wave_data.csv"
POPULATION_PATH = "../data/usvi_pop.csv"
```

2 Historical Data Window and User Inputs

```
START_DATE = "1985-01-01"
END_DATE = "2024-12-28"

MS_TO_KNOTS = 1.94384449
POP_SCALE = 1000

RISK_FREE_RATE = 0.037
AVG_PROFIT_PER_DAY = 330.0

# CHOSEN FDI THRESHOLDS
WIND_THRESHOLD_MS = 8.5
WAVE_THRESHOLD_M = 1.75

CONTRACT_START_MMDD = "01-01"
CONTRACT_END_MMDD = "12-31"

# THRESHOLD SENSITIVITY GRID
wind_thresholds = [7.5, 8.5, 10.0, 11.5, 13.0]
wave_thresholds = [1.5, 1.75, 2.0, 2.25, 2.5]

# STRIKE GRID
STRIKE_PERCENTS = np.round(np.arange(0.80, 0.951, 0.01), 2)
```

3 LOAD DATA AND SENSITIVITY ANALYSIS

```
# LOAD DATA
df_fishing = pd.read_csv(
    FISHING_PATH,
    parse_dates=["date"],
    date_format="%m/%d/%Y"
)
df_fishing = df_fishing[["date", "weekday", "n_trips", "n_csl_trips", "all_all", "csl_all"]]

df_wind = pd.read_csv(WIND_PATH, parse_dates=["valid_time"])
df_wind = df_wind[["valid_time", "u10", "v10", "fg10"]].copy()

df_wave = pd.read_csv(WAVE_PATH, parse_dates=["valid_time"])
df_wave = df_wave[["valid_time", "swh"]].copy()

df_pop = pd.read_csv(POPULATION_PATH, parse_dates=["observation_date"])
df_pop = df_pop.rename(columns={
    "observation_date": "year",
    "POPTOTVIA647NWDB": "population"
})
df_pop["year"] = df_pop["year"].dt.year

# BUILD DAILY MASTER TABLE
dates = pd.date_range(start=START_DATE, end=END_DATE, freq="D")
df_calendar = pd.DataFrame({"date": dates})

df_calendar["year"] = df_calendar["date"].dt.year
df_calendar["weekday"] = df_calendar["date"].dt.day_name()
df_calendar["is_sunday"] = (df_calendar["weekday"] == "Sunday").astype(int)

df_calendar["date"] = pd.to_datetime(df_calendar["date"], format="%m/%d/%Y")
df_fishing["date"] = pd.to_datetime(df_fishing["date"], format="%m/%d/%Y")

df_master = df_calendar.merge(df_fishing, on=["date", "weekday"], how="left")

for col in ["n_trips", "n_csl_trips", "all_all", "csl_all"]:
    df_master[col] = df_master[col].fillna(0)
```

PROCESS WIND

```

df_wind["wind_speed_ms"] = np.sqrt(df_wind["u10"]**2 + df_wind["v10"]**2)
df_wind["date"] = df_wind["valid_time"].dt.normalize()

df_wind_daily = (
    df_wind.groupby("date")
    .agg(
        max_wind_ms=("wind_speed_ms", "max"),
        mean_wind_ms=("wind_speed_ms", "mean"),
        max_gust_ms=("fg10", "max")
    )
    .reset_index()
)

```

PROCESS WAVE

```

df_wave["date"] = df_wave["valid_time"].dt.normalize()

df_wave_daily = (
    df_wave.groupby("date")
    .agg(
        max_swh_m=("swh", "max"),
        mean_swh_m=("swh", "mean")
    )
    .reset_index()
)

```

MERGE ALL

```

df_master = df_master.merge(df_wind_daily, on="date", how="left")
df_master = df_master.merge(df_wave_daily, on="date", how="left")
df_master = df_master.merge(df_pop[["year", "population"]], on="year", how="left")

df_master["csl_trips_per_1k"] = (
    (df_master["n_csl_trips"] / df_master["population"]) * POP_SCALE
)

df_master["csl_lbs_per_1k"] = (
    (df_master["csl_all"] / df_master["population"]) * POP_SCALE
)

```

```

# THRESHOLD SENSITIVITY

sensitivity_rows = []

for wnd in wind_thresholds:
    for wav in wave_thresholds:
        fdi_temp = ((df_master["max_wind_ms"] <= wnd) &
                     (df_master["max_swh_m"] <= wav)).astype(int)

        annual_temp = (
            pd.DataFrame({
                "year": df_master["year"],
                "fdi_temp": fdi_temp
            })
            .groupby("year")
            .agg(
                annual_fdi=("fdi_temp", "sum"),
                pct_fishable=("fdi_temp", lambda x: x.mean() * 100)
            )
            .reset_index()
        )

        sensitivity_rows.append({
            "wind_thresh_ms": wnd,
            "wind_thresh_kt": wnd * MS_TO_KNOTS,
            "wave_thresh_m": wav,
            "pct_fishable_total": fdi_temp.mean() * 100,
            "mean_annual_fdi": annual_temp["annual_fdi"].mean(),
            "std_annual_fdi": annual_temp["annual_fdi"].std()
        })

df_sensitivity = pd.DataFrame(sensitivity_rows)

print("\nThreshold Sensitivity Table:")
print(df_sensitivity.round(2).to_string(index=False))

```

Threshold Sensitivity Table:

wind_thresh_ms	wind_thresh_kt	wave_thresh_m	pct_fishable_total	mean_annual_fdi	std_annual_fdi
7.5	14.58	1.50	33.35	121.80	
7.5	14.58	1.75	34.56	126.20	
7.5	14.58	2.00	34.83	127.18	

7.5	14.58	2.25	34.87	127.35
7.5	14.58	2.50	34.91	127.48
8.5	16.52	1.50	52.69	192.42
8.5	16.52	1.75	56.67	206.95
8.5	16.52	2.00	57.31	209.30
8.5	16.52	2.25	57.43	209.72
8.5	16.52	2.50	57.47	209.88
10.0	19.44	1.50	62.59	228.58
10.0	19.44	1.75	80.15	292.68
10.0	19.44	2.00	85.61	312.62
10.0	19.44	2.25	86.40	315.50
10.0	19.44	2.50	86.51	315.90
11.5	22.35	1.50	62.65	228.78
11.5	22.35	1.75	81.67	298.23
11.5	22.35	2.00	91.58	334.42
11.5	22.35	2.25	96.21	351.32
11.5	22.35	2.50	97.41	355.72
13.0	25.27	1.50	62.65	228.78
13.0	25.27	1.75	81.67	298.25
13.0	25.27	2.00	91.67	334.75
13.0	25.27	2.25	96.54	352.52
13.0	25.27	2.50	98.47	359.60

```
# BUILD FINAL FDI
```

```
df_master["fdi"] = (
    (df_master["max_wind_ms"] <= WIND_THRESHOLD_MS) &
    (df_master["max_swh_m"] <= WAVE_THRESHOLD_M)
).astype(int)
```

```
print("\nSelected Thresholds:")
```

```
print(f"Wind threshold = {WIND_THRESHOLD_MS:.2f} m/s ({WIND_THRESHOLD_MS * MS_TO_KNOTS:.2f} knots)")
```

```
print(f"Wave threshold = {WAVE_THRESHOLD_M:.2f} m")
```

Selected Thresholds:

Wind threshold = 8.50 m/s (16.52 knots)

Wave threshold = 1.75 m

REMOVE FEB 29 FOR CALENDAR ALIGNMENT

```

df_prob = df_master.copy()
df_prob["date"] = pd.to_datetime(df_prob["date"])

df_prob = df_prob[
    ~((df_prob["date"].dt.month == 2) & (df_prob["date"].dt.day == 29))
].copy()

df_prob["month_day"] = df_prob["date"].dt.strftime("%m-%d")

calendar_365 = pd.date_range("2001-01-01", "2001-12-31", freq="D")
calendar_df = pd.DataFrame({
    "contract_date_dummy": calendar_365,
    "month_day": calendar_365.strftime("%m-%d"),
    "day_of_year": np.arange(1, 366)
})

```

CONTRACT WINDOW

```

start_dummy = pd.to_datetime(f"2001-{{CONTRACT_START_MMDD}}")
end_dummy = pd.to_datetime(f"2001-{{CONTRACT_END_MMDD}}")

if start_dummy <= end_dummy:
    contract_calendar = calendar_df[
        (calendar_df["contract_date_dummy"] >= start_dummy) &
        (calendar_df["contract_date_dummy"] <= end_dummy)
    ].copy()
else:
    contract_calendar = calendar_df[
        (calendar_df["contract_date_dummy"] >= start_dummy) |
        (calendar_df["contract_date_dummy"] <= end_dummy)
    ].copy()

contract_calendar = contract_calendar.reset_index(drop=True)
contract_calendar["contract_day_index"] = np.arange(1, len(contract_calendar) + 1)

contract_days = len(contract_calendar)
contract_years = contract_days / 365.0

print("\nContract Window:")
print(f"Start (MM-DD): {{CONTRACT_START_MMDD}}")
print(f"End (MM-DD): {{CONTRACT_END_MMDD}}")

```



```
print(f"Contract length: {contract_days} days")
print(f"Contract length: {contract_years:.6f} years")
```

```
Contract Window:
Start (MM-DD): 01-01
End (MM-DD): 12-31
Contract length: 365 days
Contract length: 1.000000 years
```

DAILY FISHABLE PROBABILITY FOR CONTRACT DAYS ONLY

```
daily_prob_all = (
    df_prob.groupby("month_day")
    .agg(
        p_fishable=("fdi", "mean"),
        fishable_count=("fdi", "sum"),
        n_years=("fdi", "count")
    )
    .reset_index()
)

daily_prob_contract = contract_calendar.merge(
    daily_prob_all,
    on="month_day",
    how="left"
)

if daily_prob_contract["p_fishable"].isna().any():
    raise ValueError("Some contract dates have missing fishable probabilities.")

expected_fishable_days_contract = daily_prob_contract["p_fishable"].sum()

print("\nContract Probability Summary:")
print(f"Expected fishable days during contract = {expected_fishable_days_contract:.2f}")

# CONTRACT-YEAR FDI FOR HISTORICAL YEARS

contract_month_days = set(contract_calendar["month_day"])

df_contract_only = df_prob[df_prob["month_day"].isin(contract_month_days)].copy()
```

```

annual_contract_fdi = (
    df_contract_only.groupby("year")
    .agg(
        contract_fdi=("fdi", "sum"),
        contract_days=("fdi", "count"),
        pct_fishable=("fdi", lambda x: x.mean() * 100)
    )
    .reset_index()
)

mean_contract_fdi = annual_contract_fdi["contract_fdi"].mean()
std_contract_fdi = annual_contract_fdi["contract_fdi"].std()

print("\nContract FDI Summary:")
print(f"Mean contract FDI = {mean_contract_fdi:.2f} days")
print(f"Std contract FDI = {std_contract_fdi:.2f} days")
print(f"Average profit/day = ${AVG_PROFIT_PER_DAY:,.2f}")

```

Contract Probability Summary:
 Expected fishable days during contract = 206.80

Contract FDI Summary:
 Mean contract FDI = 206.78 days
 Std contract FDI = 23.94 days
 Average profit/day = \$330.00

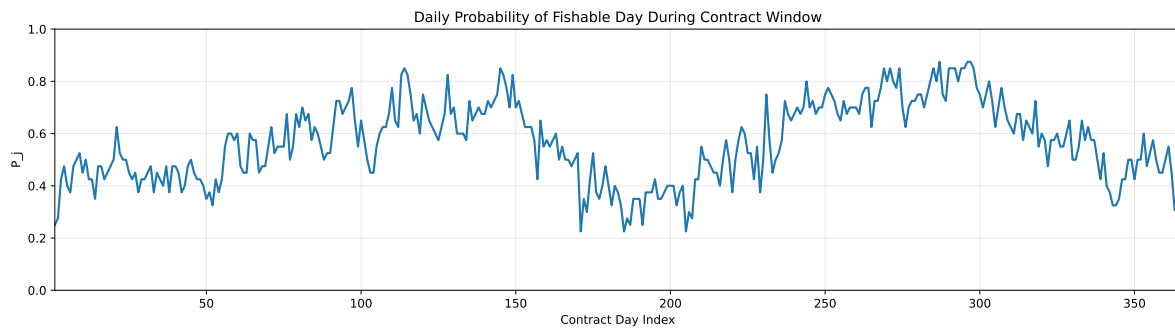
PLOT DAILY FISHABLE PROBABILITY

```

plt.figure(figsize=(14, 4))
plt.plot(
    daily_prob_contract["contract_day_index"],
    daily_prob_contract["p_fishable"],
    linewidth=1.8
)
plt.title("Daily Probability of Fishable Day During Contract Window")
plt.xlabel("Contract Day Index")
plt.ylabel("P_j")
plt.xlim(1, contract_days)
plt.ylim(0, 1)
plt.grid(True, alpha=0.3)

```

```
plt.tight_layout()
plt.show()
```



POISSON-BINOMIAL PMF

```
=====

p = daily_prob_contract["p_fishable"].to_numpy()
n = len(p)

pmf = np.zeros(n + 1)
pmf[0] = 1.0

for pj in p:
    new_pmf = np.zeros(n + 1)
    new_pmf[0] = pmf[0] * (1 - pj)
    for k in range(1, n + 1):
        new_pmf[k] = pmf[k] * (1 - pj) + pmf[k - 1] * pj
    pmf = new_pmf

discount_factor = np.exp(-RISK_FREE_RATE * contract_years)

# MULTI-STRIKE PRICING TABLE || Trigger rule: payout when N < K

pricing_rows = []

for strike_percent in STRIKE_PERCENTS:
    strike_days = int(round(mean_contract_fdi * strike_percent))

    # Fixed payout size based on strike shortfall from mean
    loss_amount = max(mean_contract_fdi - strike_days, 0) * AVG_PROFIT_PER_DAY
```

```

# Correct trigger rule: fishable days strictly below strike
#  $P(N < K) = \text{sum from } 0 \text{ to } K-1$ 
if strike_days <= 0:
    trigger_probability = 0.0
else:
    trigger_probability = pmf[:strike_days].sum()

weather_derivative_price = discount_factor * loss_amount * trigger_probability

pricing_rows.append({
    "strike_percent": strike_percent * 100,
    "strike_days": strike_days,
    "fixed_payout_L": loss_amount,
    "trigger_probability_P_N_less_K": trigger_probability,
    "weather_derivative_price": weather_derivative_price
})

pricing_table = pd.DataFrame(pricing_rows)

print("\nPricing Table for Strike Percent 80% to 95%:")
print(
    pricing_table.round({
        "strike_percent": 0,
        "strike_days": 0,
        "fixed_payout_L": 2,
        "trigger_probability_P_N_less_K": 6,
        "weather_derivative_price": 2
    }).to_string(index=False)
)

# PLOT - STRIKE % VS PRICE

plt.figure(figsize=(10, 5))
plt.plot(
    pricing_table["strike_percent"],
    pricing_table["weather_derivative_price"],
    marker="o",
    linewidth=1.8
)
plt.title("Weather Derivative Price vs Strike Percentage")

```

```

plt.xlabel("Strike Percentage of Mean Fishable Days")
plt.ylabel("Weather Derivative Price ($)")

plt.xticks(STRIKE_PERCENTS * 100)

plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# PLOT - STRIKE % VS TRIGGER PROBABILITY

plt.figure(figsize=(10, 5))
plt.plot(
    pricing_table["strike_percent"],
    pricing_table["trigger_probability_P_N_less_K"],
    marker="o",
    linewidth=1.8
)
plt.title("Trigger Probability vs Strike Percentage")
plt.xlabel("Strike Percentage of Mean Fishable Days")
plt.ylabel("Trigger Probability")

plt.xticks(STRIKE_PERCENTS * 100)

plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

Pricing Table for Strike Percent 80% to 95%:

strike_percent	strike_days	fixed_payout_L	trigger_probability_P_N_less_K	weather_derivativ
80.0	165	13785.75	0.000002	
81.0	167	13125.75	0.000004	
82.0	170	12135.75	0.000019	
83.0	172	11475.75	0.000049	
84.0	174	10815.75	0.000119	
85.0	176	10155.75	0.000276	
86.0	178	9495.75	0.000610	
87.0	180	8835.75	0.001289	
88.0	182	8175.75	0.002605	
89.0	184	7515.75	0.005032	
90.0	186	6855.75	0.009303	

91.0	188	6195.75	0.016460
92.0	190	5535.75	0.027895
93.0	192	4875.75	0.045302
94.0	194	4215.75	0.070555
95.0	196	3555.75	0.105465

